

REQUEST FOR PROPOSAL: Store Low Voltage Cabling



Purpose

The purpose of this Request for Proposal (RFP) is to invite vendors to submit their proposal for the furnishing, installation, and testing of low voltage copper and optical fiber cabling system throughout our stores being built and renovated for Lidl US.

Company Background

Lidl is one of the leading global retailers and currently operates nearly 10,000 stores in 29 countries throughout Europe and the United States. Lidl has a simple philosophy: we are focused on delivering customers the highest quality products at the lowest possible prices in convenient locations.

Lidl markets offer customers fresh meats and poultry, produce and bakery items, as well as a wide selection of household goods and wines, at the lowest possible prices.

General Requirements

The successful Contractors shall furnish all labor, materials, equipment, testing, and supervision to provide the required low voltage cabling system in accordance with the general requirements, Scope of Work (SOW), and technical specifications presented in this RFP document and any supplements thereto.

Statement Of Work

Scope Specifications

Work Area Outlets

Contractor (low voltage contractor) will provide and install a Category 6A F/UTP copper cable solution for the CCTV, wireless, and data work area outlets (WAOs) located throughout the store. Cable will be R&M Category 6A Plenum F/UTP copper cable for data, CCTV, and wireless (R859102), and Category 6A Shielded F/UTP Modular Jacks will be used for termination at the WAO's and in the IT Room.

The contractor will furnish and install these cables from the IT Room located in the front of the store to the associated WAO's as noted on the provided floor plan (Appendix A).

There are no j-hooks allowed to be used. The cables must be laid either on a cable tray or within an EMT conduit. The cable tray and EMT conduit along with data outlet boxes, bushings for the conduit, and pull string will be installed by the General Contractor (GC). It is a responsibility of Contractor to report to the GC if any of the required cable trays and conduit marked on the plans are missing or are in unacceptable condition which will not allow the proper installation of the cable. The cable tray size will vary in different sections of the store depending on how many cables will be installed in those areas.

WAPS

There are fifteen (15) WAOs designed for Wireless Access Points (WAP). These locations are marked on Appendix A as AP01 – AP15 along with highlighted by a YELLOW dot. There will be a 4" electrical j-box and conduit with pull string provided by the GC for each of these locations. The Category 6A F-UTP cable will be terminated with a Category 6A Shielded female jack in the box without a face plate at the WAP location. Lidl IT will provide all WAPS and Contractor will be responsible for mounting each WAP at their designated locations.

CCTV

There are twenty-seven (27) CCTV camera locations on Appendix -A marked by a camera icon along with highlighted by a PURPLE dot on drawing T-201. Each location will receive one (1) Category 6A F-UTP cable which will be terminated with a Category 6A Shielded jack in a single port faceplate mounted on a 4" electrical box. Lidl will provide and mount all cameras at their designated locations.

Data Outlets

There are nineteen (19) WAO's which require two (2) Category 6A F/UTP cables pulled to the IT Room and are marked by a RED dot on Appendix A Drawing T-101. Each of the dual outlets will require a 2 Port faceplate to be installed at the workstation. Along with the nineteen (19) dual WAO locations, there are two (2) WAO's which will require four (4) Category 6A F/UTP cables. These locations are the Manager's Office (Office 108) and the BMS outlet located in the Technical Room (111). These WAO's will be pulled back to the IT Room and are marked with Green dots on Appendix A. These locations will require a 4-Port faceplate to be installed.

Please note of the nineteen (19) dual WAO, there are three (3) WAO's located in the bakery which will require a 2-Port Stainless Steel faceplate.

There are eleven (11) WAO's which require four (4) cables pulled back to the IT Room and are marked by a GREEN dot on Appendix A. Each of these eleven (11) WAO's are for checkouts and self-checkouts (SCO's) and will be terminated on 4-port surface mount box within the cable tray at the station side. There will be a total of forty-four (44) cables terminated in the cable tray for the checkouts. Lidl will provide Contractor with 65 ft patch cords which will be pre-terminated with RJ45 mod ends on one side and unterminated on the other. For each checkout, Contractor will have to plug four (4) of the pre-terminated side patch cords into the 4-port surface mount box inside the cable tray and run the unterminated sides down the power pole into the checkout where Contractor will terminate onto another Lidl provide 4-port surface mount box located inside a secure data box. Contractor will have to mount the secure data box within the checkout at its designated location.

Contractor will provide and install two (2) Category 5E plenum copper cables from the demarc location to the Fire Alarm Control Panel (FACP) and terminate on RJ11 in the FACP. These cables shall be labeled according to the information shown in Appendix B. Contractor will also be required to have a butt set to test the POTS lines at the FACP once the Service Provider installs the lines at the demarc.

General Cabling Installation Requirements

Contractor will ensure all cabling is installed end to end without splices and under no circumstances shall the copper cable length between the IT Room and the furthest WAO exceed 295 feet. Contractor must ensure that all cabling is routed and dressed into the cabinets in a neat and aesthetically pleasing fashion. It must be ensured the bend radius of all cabling does not exceed the limit of 4 times the outer diameter of the cable. Each cable bundle will be neatly routed, dressed, and terminated to their respective block or panel for termination. All cables must be managed with the proper cable management hardware. The Contractor must ensure when each cable is terminated, the cable jacket will be cut back as close to the termination point as possible, and the pair untwist does not exceed ½" at any time. The cable jacket must be maintained as close as possible to the termination point, and each cable must be clearly labeled on the cable jacket behind the patch panel or block at a location that can be viewed without removing the bundle support ties. All cables in the IT Room shall be bundled using white Velcro-type ties. Nylon cable ties shall not be used. Cables showing outer jacket abrasion will be replaced at the option of Lidl. The replacement cost will be the responsibility of Contractor. All cables must be properly labeled with a machine-generated wire wrap label, and according to those standards set forth by ANSI/TIA/EIA-606 standards for cable plant administration and in Appendix B.

Maintenance loops or slack are required for cabling and shall not be any less than 10-ft (3m) at the IT Room cabinets, along with no less than a 2-ft (60cm) at the WAO's.

Maintenance loops shall also be provided on the cable tray above the IT Room cabinets. Slack at the WAO's shall not be placed in the outlet box, rather installed at the nearest practical point of the conduit at the cable tray.

IT Room: Cabinet Installation

Contractor will install the cabinets as defined by ANSI/TIA/EIA-568B and BICSI standards, along with positioning the layout in accordance to Appendix C. Cabinets within the IT Room shall be placed side by side with Cabinet #1 always being located next to the wall. After final stabilization of the cabinets, the power plugs of the PDUs must be plugged into the receptacles installed above the cabinets. When facing the rear of the cabinets the PDU's located on the left will plug into the UPS receptacles and the PDU's located on the right will plug into the utility power receptacles. The PDU's will be labeled indicating whether it is plugged into the UPS or utility power receptacles. Once the PDU's are plugged in, Contractor is required to nicely dress the power cords and not let them hang.

Once the cabinets are delivered to site, Contractor will be required to move cabinets from loading dock into IT Room using nylon pads to ensure not to scratch the floors.

All cabinets in the IT Room shall be properly ground by Contractor per the grounding specification found in this document Appendix E.

Please reference Appendix C for proper rack elevations and Contractor installation requirements. The cabinets will be delivered to the site preloaded and Contractor is required to install the copper and fiber patch panels. Each patch panel should be installed with a rear cable management bar to ensure that the proper bend radius and tension is applied to each cable. Along with installing the patch panels,

Contractor will be responsible to installing the 19" toolless blanking panels within Cabinet #2 for all unused RU which are shipped with the cabinets.

The GC will be installing the basket tray overhead within the IT Room for proper horizontal cable management, however Contractor must furnish and install the appropriately sized cable management waterfalls from the basket tray for cables entering the cabinet.

Closet Cable Interconnections:

Please reference Appendix D for all specified copper cable interconnections from cabinet to cabinet in the IT Room.

Closet Patching:

Contractor will not be responsible for any patching within the closets.

Closet Racking and Stacking:

Contractor will not be responsible for any racking and stacking of any equipment.

GPS Antenna

Lidl will furnish a GPS antenna and Contractor will accept the delivery once onsite. Contractor will mount the GPS antenna at a roof location which will be coordinated onsite with the GC. Once Contractor installs the antenna, they must run the coax cable (which is included with the antenna) from the roof down to the Bakery cabinet through a conduit with pull string installed by the GC. The Bakery cabinet location will be coordinated once onsite with Lidl. Contractor is responsible for connecting the coax cable at the antenna and leave coiled at the Bakery cabinet.

Grounding

All telecommunications grounding shall comply with all applicable laws, regulations, standards, codes, and industry practices, as well as those guidelines detailed in EIA/TIA-607 – Commercial Building Grounding and Bonding Requirements for Telecommunications.

Contractor shall bond all the cabinets and patch panels to the ground bus bar (provided by GC) located in the IT Room.

The GC will provide a grounding busbar in the IT Room. Contractor shall remove the surface paint on the cabinets, so the conductor can be bonded directly to the surface metal. An insulated #6 AWG copper wire shall be used for grounding all communications cable and cabinets. The Contractor shall furnish and install a horizontal bus bar (Panduit RGRB19CN Grounding Busbar Kit) in each IT Room cabinet. An insulated #12 AWG copper wire shall be used for grounding all communications panels (Appendix E). The installation of fault protection grounding shall conform to the requirements of Article 250 of the National Electrical Code.

Each ground cable shall be labeled at both ends with the FROM and TO information so that each ground wire can be easily identified.

Labeling

All WAO's and patch panels need to be labeled according to the information shown in Appendix B.

All labeling information shall be recorded on the as-built drawings and all test documents shall reflect the appropriate labeling scheme. Self-laminating wire wrap labels will be used on cable jackets, appropriately sized to the OD of the cable, and placed within view at the termination point on each end. For every outlet, patch panel, and wiring block, each label shall be installed in the space provided on that device.

Cables shall be identified by a self-adhesive wire wrap labels in accordance with Appendix B and ANSI/TIA-606-A. The cable label shall be applied to the cable within 12" behind the faceplate and patch panel.

All cables, patch panels, face plates, and blocks must be labeled on both ends with location ID and cable type. Station location wiring identification should be based on industry standard identification schemes allowing any given wiring pair label attachment to be traced from its origin to its final point of termination.

Testing

Contractor shall test 100% of the terminated copper pairs and 100% of the terminated fiber strands and all cables must meet or exceed the guidelines set out in the ANSI/TIA/EIA T568 standards for basic link test. Contractor must use Fluke DTX or DSX model testers. Contractor must set the tester to test to the specific cable type and manufacture they are testing. Contractor shall provide a complete set of test results covering each tested link. The test results shall be saved in the tester's proprietary graphical format as well as PDF format and provided electronically either via email, FTP site, or CD. Either format shall include the tester manufacturer's proprietary test report viewer. The test results must be named with a standard naming convention previously described in this document. Each test must have at least the following test performed for each cable:

- 1) Cable Length
- 2) Near-end cross talk (NEXT)
- 3) Equal level far-end (ELFEXT)
- 4) Propagation delay
- 5) Wire Map
- 6) Attenuation/Insertion Loss

Contractor shall repair or replace any link that does not pass all tests. Marginal passing results do not qualify as a passing test.

The test result documentation must include the following:

- 1.) Date of each test
- 2.) Test results summary page and individual results
- 3.) Description of test equipment used
- 4.) Date of latest calibration of test equipment. Calibration date must not be older than one (1) year

Test results shall be provided within two (2) weeks of completion of the project and prior to final payment to Contractor.

As-Built Documentation

Contractor shall provide all installed cable locations complete with as-built location numbers within 2 weeks of completion of the project and prior to final payment to Contractor. Contractor also will be required to provide an as built drawing mounted inside of the IT Room upon completion of the project. The drawing shall be framed and no small than 24x30.

Warranty:

Contractor shall provide a minimum 25 year parts and labor manufacturer's warranty. All proposals regarding the cable under this RFP shall reference coverage against defects in material and workmanship. All parts, equipment, cables, connectors and other materials and workmanship furnished by the Contractor shall include the manufacturer's warranty for replacement from the date of acceptance of Contractor's work by Lidl US.

Contractor further agrees to replace all components which fail or do not perform according to the manufacturer's specifications during the warranty period at no cost to Lidl US.

Acceptance

Lidl will grant cabling and systems acceptance to Contractor following a successful systems cutover by Lidl POS Team. This acceptance will also be predicated upon the Contractor providing Lidl with the following items:

- 1.) Certification testing documentation
- 2.) As-built documentation
- 3.) Correction and resolution of any punch list items cited by Lidl
- 4.) Provision of official warranty information
- 5.) Final pictures

Non-Prototypical Sites

Lidl now has a revised model of expanding our footprint where we not only build our sites from the ground up, rather we are taking on existing spaces as well. With this new style of expansion, all our sites will no longer all be cookie cutter in the sense of building orientation. However, the number of outlet locations will remain the same. With the varying building orientations, we will have sites which we will need to get a revised priced due to a larger footprinted store which will require an SDR (Sub Distribution Rack) to accommodate the longer cable runs.

Lidl will provide Contractor with the Non-Prototypical site-specific drawing and ask for updated pricing to be turned around within 1 week. We will evaluate the proposals and award the site to the winning Contractor. All requirements of the RFP stated above still apply to the Non-Prototypical sites. Below are two (2) additional requirements which will be added to the Non-Prototypical sites.

Backbone Cabling (Non-Prototypical Sites)

Contractor will provide and install one (1) 24-strand Multi-Mode fiber backbone from the Server Room Cabinet #2 to the SDR. Contractor will also provide and install twelve (12) Category 6A F/UTP cables from the Server Room Cabinet #2 to the SDR. An updated port allocation sheet (Appendix D) and cabinet layout (Appendix C) for patch panel location purposes will be provided along with the drawings.

SDR Area: Cabinet Installation (Non-Prototypical Sites)

The GC will mount the SDR and securely fasten it to the plywood, utilizing the appropriate hardware for the applicable wall. There is a preinstalled PDU in the rear of the SDR and Contractor will need to plug this PDU into the UPS receptacle on the wall and the PDU in the front will plug into the utility power receptacle. Each PDU will come labeled indicating whether it is plugged into the UPS or utility power receptacles.

The SDR cabinet shall be properly ground by Contractor per the grounding specification found in this document (Appendix E).

Please reference Appendix C for proper rack elevations and Contractor installation requirements. The cabinets will be delivered to the site preloaded and Contractor is required to install the copper and fiber patch panels. Each patch panel should be installed with a rear cable management bar to ensure that the proper bend radius and tension is applied to each cable.

The GC will be installing the basket tray overhead the SDR Area for proper horizontal cable management, however Contractor must furnish and install the appropriately sized cable management waterfalls from the basket tray for cables entering the SDR.

General Specifications

Ordering of Materials/Services

Outside of the material list (Appendix F) ordered by Contractor, Contractor will also be responsible for ordering, accepting delivery, inspecting, and inventorying all the following Lidl furnished equipment:

- 1) Telecommunication Cabinets— A point of contact (POC) from Vertiv will be provided to Contractor by Lidl. Contractor must contact the Vertiv POC once they are provided their contact information. This will allow Contractor to get delivery timeframes and schedule delivery in a timely manner for that store's project. Once a delivery date is agreed upon, Contractor must submit order via email to the Vertiv POC to document the order.
- 2) WAPS – Contractor contact Lidl IT and schedule when they will need the WAPS onsite.
- 3) Patch Cords and Secure Data Boxes – Contractor will contact Lidl IT and schedule when they will need the 65ft register patch cords, WAP patch cords, and secure boxes for the registers onsite.

Contractor will not have an obligation to pay for these materials/services as they will be prepaid by Lidl.

Reporting Requirements

Contractor will be responsible to attend a daily conference call with Lidl IT to discuss progress of the active projects. Contractor will also be responsible for submitting a weekly tracking report to chronical their installation progress and onsite issues and or obstacles. A sample weekly status report is to be submitted with this bid.

At the completion of the job, Contractor is to provide final pictures of the following:

- a.) Each workstation jack installed, clearly showing the mechanical labeling
- b.) Patch panel terminations showing the mechanical labeling for the wires ran
- c.) Each of the completed terminations inside the basket tray for the checkouts
- d.) Each WAP location
- e.) GPS Antenna Installed in its final location

Lift Requirements

Cabling our sites will require a lift to be used. When operating a lift onsite, it is a requirement the lift be outfitted with diapers and booties.

Low Voltage Permits

It is Contractor responsibility to obtain all required permits and inspections. Contractor is responsible for coordinating their own inspections so they do not impact the General Contractor's final inspections. Contractor is to provide a copy of their Low Voltage Permit within the 1st week onsite and a copy of their "Passed" final inspection by the end of the last week onsite.

Safety

Contractor is always expected to work in a safe and professional manner. Contractor is expected to follow all applicable safety rules and regulations at each site, which includes but is not limited to OSHA, General Contractor, Lidl, and Low Voltage Contractor. All techs working onsite must wear proper PPE the entirety of the job.

Working Hours

Contractor shall only work during the General Contractor's normal hours for each job site, unless previously agreed upon with Lidl and the General Contractor.

Work Area

At the end of each day, Contractor shall remove all debris and secure all unused materials.

Proposal Submission Procedure

Duration of Scope Of Work (SOW)

The SOW described in this RFP must be completed in no more than (4) weeks per site. However, the expectation is for Contractor to be onsite daily and complete the SOW as soon as possible. The four (4) week window is the timeframe allotted for completing the SOW, construction delays, and punch list completion. Contractor is expected to price to the SOW and not just the 4 week duration.

Evaluation Criteria

All proposals will be evaluated based on, but not limited to, the following criteria:

1. Professional quality, presentation, and clarity of the proposal
2. Relevant experience and specific examples provided for exceptional service
3. Competitive price structure
4. Best alignment with the Lidl US IT requirements and geographical coverage for the following states: South Carolina, North Carolina, Virginia, Maryland, Delaware, New Jersey, New York, Georgia, and Pennsylvania.

Proposal Content

Proposals will be accepted from Contractor whom demonstrated experience and competency in the field of low voltage cabling. Responses should include the following information to be considered:

Vendor Information

- **Corporate Overview:** company legal name, number of employees, relevant industry experience, annual report if available
- **Services & Products:** description of relevant products and/or services offered
- **Client References:** provide at least three (3) recommendations from related industry clients your organization has served or is currently serving if possible
- **Differentiators:** Describe how your company distinguishes itself from others and why our selection of your company is the best decision
- **Professional liability insurance:** Provide proof of insurance
- **Quality control policy**
- **Statement indicating financial stability**

Estimated Budget & Resources Required

All vendors should provide a detailed breakdown of costs by filling out the included Bill Of Materials (BOM) – Appendix F, particularly referencing the scope of work in this RFP.

All proposals should include a project schedule and breakdown of work, which will identify timelines, key milestones, project phases or other details. The completion of each SOW must always be limited to 4 weeks, unless previously notified by Lidl.